

Blue Origin Panel Interview Preparation Guide – Updated Expanded Edition

Data Acquisition and Controls Engineer II – New Glenn, R66427

Expanded Technical and Behavioral Study Notes

July 13, 2026

Abstract

This expanded edition explains the topics previously listed as preparation items. It is written as both an interview guide and a compact technical reference. It covers sensor selection, signal conditioning, ADCs, sampling, grounding, controls, safety, Python software architecture, NI/DAQmx concepts, data processing, databases, electrical troubleshooting, commissioning, system design, and behavioral interview answers. Examples, equations, and figures are included where they help explain the engineering reasoning.

Core interview message

“I can translate test objectives into a safe, traceable, and maintainable measurement-and-control system—from sensor selection and signal conditioning through acquisition, automated control, data validation, post-processing, and engineering review.”

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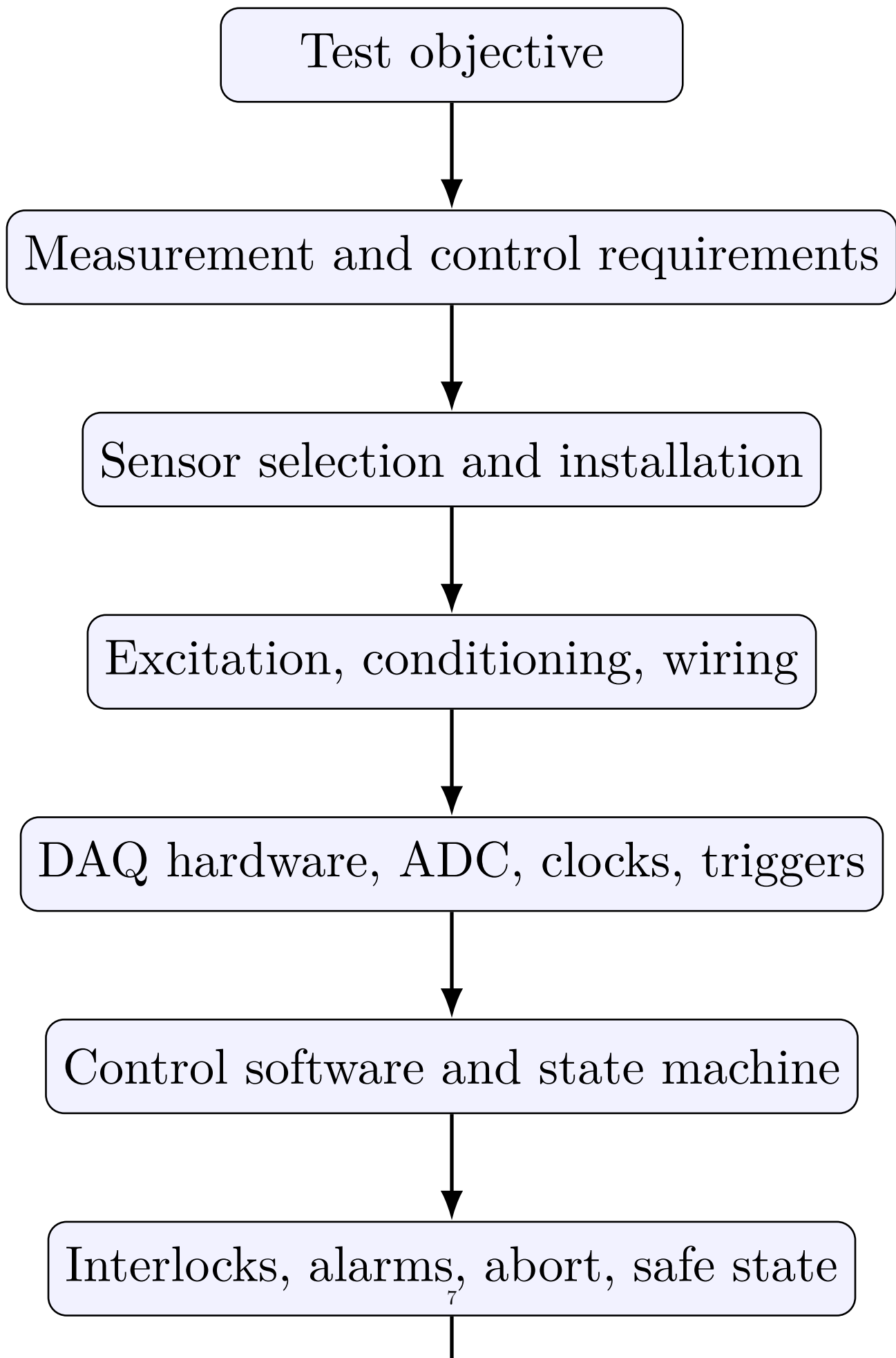
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1 How to Think About the Role

1.1 The end-to-end test system

A Data Acquisition and Controls Engineer does not only connect sensors or write scripts. The engineer owns the chain by which a physical event becomes trusted engineering evidence. A complete answer should connect hardware, software, safety, data integrity, and review.



1.2 What an Engineer II should demonstrate

At this level, the panel will likely expect:

- Independent execution on scoped systems.
- Ability to identify incomplete requirements.
- Defensible tradeoffs involving safety, data quality, schedule, cost, and maintainability.
- Troubleshooting across mechanical, electrical, controls, software, and data boundaries.
- Clear communication to test operations, design engineering, controls, analysis, and management.
- Respect for formal reviews, calibration, configuration control, and safety authority.

2 Sensor Selection: Detailed Explanations

2.1 Measurement range

The measurement range is the minimum and maximum value the sensor can measure within its specified performance. Selecting range is a tradeoff:

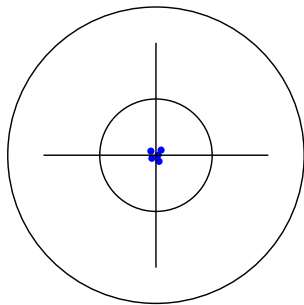
- Too small: the sensor may saturate or be damaged during transient or fault conditions.
- Too large: the useful signal occupies a small part of the range, reducing practical resolution.

Pressure range example

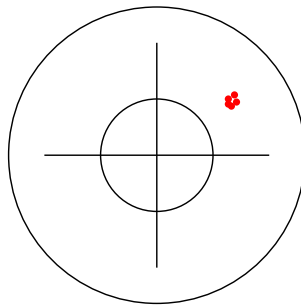
Nominal pressure is 2000 ψ , but transients may reach 2600 ψ . A 2000 ψ transducer is inappropriate because it may saturate during credible operation. A 3000 ψ transducer may be reasonable if proof pressure, burst pressure, accuracy, and resolution are acceptable.

2.2 Accuracy versus precision

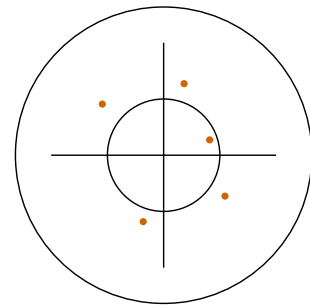
Accuracy is closeness to the true value. **Precision** is repeatability or tight clustering of repeated readings. A precise sensor can be inaccurate if it has an offset; an accurate system should be both calibrated and repeatable enough for the decision being made.



Accurate and precise



Precise, not accurate



Neither

2.3 Sensitivity

Sensitivity is output change per input change. For example, a pressure sensor may output 10 mV/V at full scale or 5 V over a 0 to 3000 ψ range. High sensitivity is useful when small changes must be resolved, but it does not guarantee accuracy.

2.4 Resolution

Resolution is the smallest distinguishable increment. It is affected by the sensor, signal conditioner, ADC span, noise, and digital representation.

Resolution example

If a 0 to 3000 ψ transducer is digitized over a 0 to 10 V range using a 16-bit ADC, the ideal pressure step is approximately

$$\Delta P = \frac{3000 \psi}{2^{16}} \approx 0.046 \psi.$$

This is ideal resolution, not total accuracy. Noise and calibration can make the effective resolution worse.

2.5 Linearity

Linearity describes how closely sensor output follows a straight-line relationship with input. Nonlinearity is often specified as a percentage of full scale. A nonlinear sensor can still be useful if a calibration curve or linearization model is applied.

2.6 Hysteresis

Hysteresis means the output depends on whether the input is approached from above or below. For example, a pressure transducer may read slightly differently at 1000 ψ during pressurization than during depressurization. This matters when acceptance criteria are tight or when load/unload cycles are evaluated.

2.7 Repeatability

Repeatability is the ability to return to the same reading under the same conditions. In test operations, repeatability affects confidence that trends are caused by hardware behavior rather than measurement instability.

2.8 Drift

Drift is slow change in sensor output over time, temperature, or usage. Zero drift shifts the output at no input; span drift changes sensitivity. Drift is managed with warmup periods, calibration intervals, temperature compensation, and pre/post-test zero checks.

2.9 Bandwidth and dynamic response

Bandwidth is the frequency range over which the sensor accurately follows input. A slow sensor can attenuate or phase-shift fast transients.

For a first-order sensor with time constant τ , the approximate cutoff frequency is:

$$f_c = \frac{1}{2\pi\tau}.$$

Dynamic response example

If a temperature probe has $\tau = 0.5$ s, then

$$f_c \approx \frac{1}{2\pi(0.5)} \approx 0.32 \text{ Hz}.$$

It is suitable for slow thermal trends but not for fast thermal transients.

2.10 Environmental rating

Sensors must survive temperature, vibration, humidity, vacuum, chemical exposure, pressure media, electromagnetic environment, and mechanical shock. Environmental compatibility is especially important in propulsion and pressurized-fluid testing.

2.11 Calibration

Calibration compares a sensor chain to a known standard. Good calibration practice includes:

- Traceability to a standard.
- Calibration date and due date.
- As-found and as-left data.
- Calibration coefficients or curves.
- Uncertainty statement.
- Configuration control so the correct coefficients are used in software.

2.12 Over-range protection

Over-range protection prevents sensor damage or unsafe behavior if input exceeds normal range. It may include mechanical stops, relief valves, snubbers, input clamps, fuses, current limiting, proof pressure margins, or software alarms.

2.13 Mounting effects and sensor loading

The sensor can disturb the system being measured. Examples:

- A pressure tap with long tubing can introduce resonance or damping.
- A thermocouple bead with large mass can slow response.
- A voltmeter with low input impedance can load a circuit.
- A load cell installed with side loads can produce erroneous readings.

3 Important Sensor Types

3.1 Pressure transducers

3.1.1 Absolute, gauge, sealed gauge, and differential

- **Absolute pressure** is measured relative to vacuum.
- **Gauge pressure** is measured relative to local atmosphere.
- **Sealed gauge** is referenced to a sealed reference pressure.
- **Differential pressure** measures the difference between two ports.

$$P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}}.$$

3.1.2 Voltage and current outputs

Common outputs include 0 to 5 V, 0 to 10 V, millivolt bridge outputs, and 4 to 20 mA. Current loops are often preferred for long cables and noisy environments.

3.1.3 Proof and burst pressure

Proof pressure is the pressure the device can withstand without permanent performance change. Burst pressure is the pressure at which containment may fail. A credible transient must remain below appropriate limits with margin.

3.1.4 Tubing and mounting effects

Pressure tubing adds volume, resistance, and compliance. It can act as a low-pass filter or resonator. If the measurement objective is fast transient pressure, mount the sensor close to the source and verify dynamic response.

3.2 Thermocouples

3.2.1 Seebeck effect

A thermocouple generates a voltage because two dissimilar metals form junctions at different temperatures. The output is small, often tens of microvolts per degree Celsius.

3.2.2 Cold-junction compensation

Thermocouple voltage depends on the temperature difference between the sensing junction and the reference junction. Cold-junction compensation measures the reference junction temperature and mathematically compensates the reading.

$$T_{\text{hot}} = f^{-1}(V_{\text{TC}} + f(T_{\text{cold}})).$$

3.2.3 Grounded versus ungrounded junctions

Grounded thermocouples have faster response but can create ground-loop or common-mode problems. Ungrounded thermocouples are slower but electrically isolated from the sheath.

3.3 RTDs

An RTD changes resistance with temperature. Platinum RTDs are common because they are stable and accurate.

$$R(T) \approx R_0(1 + \alpha T)$$

near a limited temperature range, where R_0 is resistance at 0°C and α is the temperature coefficient.

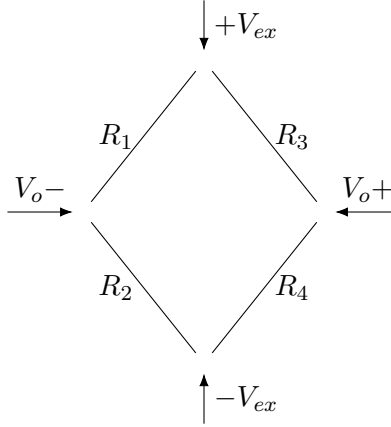
3.3.1 Two-, three-, and four-wire RTDs

- **Two-wire:** simplest, but lead resistance adds error.
- **Three-wire:** compensates lead resistance if lead wires are matched.
- **Four-wire:** best accuracy; separate excitation and sense leads.

3.4 Load cells and strain gauges

3.4.1 Wheatstone bridge

Strain gauges often use a Wheatstone bridge. A full bridge uses four active arms; a half bridge uses two; a quarter bridge uses one active gauge and completion resistors.



For a balanced bridge,

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}, \quad V_o = 0.$$

Strain changes resistance and produces a small differential output.

3.4.2 mV/V sensitivity

A load cell rated 2 mV/V produces 2 mV per volt of excitation at full scale. With 10 V excitation, full-scale output is:

$$V_{FS} = 2 \frac{\text{mV}}{\text{V}} \times 10 \text{ V} = 20 \text{ mV}.$$

This small signal requires differential low-noise amplification.

3.4.3 Shunt calibration

Shunt calibration places a known resistor across one bridge arm to simulate a known strain/load. It checks wiring, excitation, bridge completion, and scaling, but it does not replace mechanical calibration of the full load path.

4 Signal Conditioning: Expanded Explanations

4.1 Amplification

Amplification increases signal amplitude before digitization. It is useful when the sensor output is small compared with ADC input span.

$$V_{\text{out}} = G V_{\text{in}},$$

where G is gain.

Amplifying a load-cell signal

A load cell produces 0 to 20 mV. If the ADC range is 0 to 10 V, the signal uses only 0.2% of the ADC span. A gain of 250 maps 20 mV to 5 V, greatly improving usable resolution. The amplifier must not saturate during overloads.

4.2 Attenuation

Attenuation reduces signal amplitude so it fits within the measurement input range. It is common for high-voltage measurements.

For a resistor divider:

$$V_{\text{out}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}.$$

Voltage divider example

To measure 100 V with a 10 V ADC, use a 10:1 divider. If $R_1 = 90 \text{ k}\Omega$ and $R_2 = 10 \text{ k}\Omega$,

$$V_{\text{out}} = 100 \frac{10}{90 + 10} = 10 \text{ V}.$$

Check resistor power, tolerance, temperature coefficient, and isolation.

4.3 Filtering

Filtering removes unwanted frequency content. Filters can be analog or digital. Analog filtering before the ADC is required for anti-alias protection.

4.3.1 Low-pass filter

A first-order RC low-pass filter has cutoff:

$$f_c = \frac{1}{2\pi RC}.$$

Signals below f_c mostly pass; signals above f_c are attenuated.

4.3.2 High-pass, band-pass, and notch filters

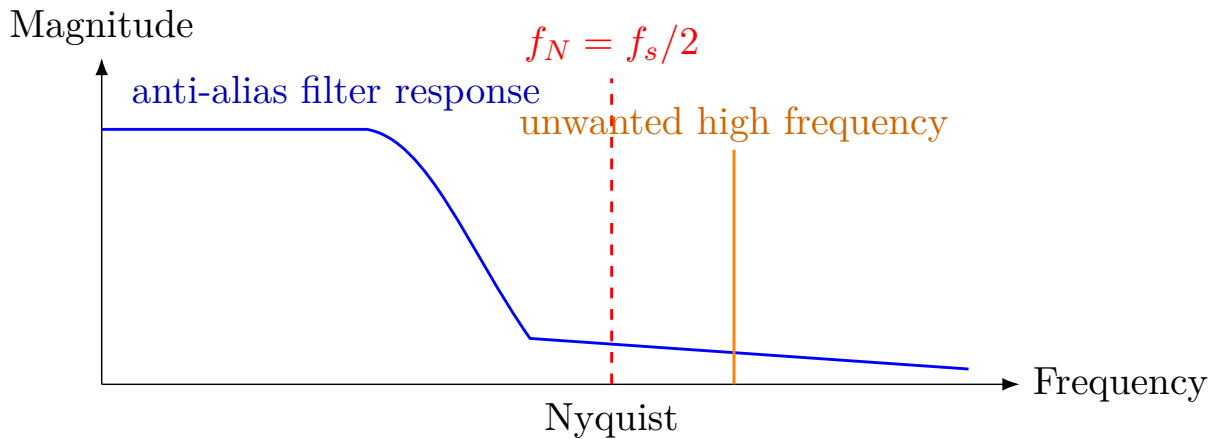
- **High-pass:** removes slow drift or DC offset.
- **Band-pass:** isolates a frequency band.
- **Notch:** rejects a narrow frequency, such as 60 Hz interference.

4.4 Anti-alias filtering

Aliasing occurs when frequency content above half the sample rate appears as a false lower-frequency signal after sampling. If sample rate is f_s , the Nyquist frequency is:

$$f_N = \frac{f_s}{2}.$$

Analog content above f_N must be attenuated before digitization.



4.5 Isolation

Isolation breaks direct electrical connection between two parts of a system while allowing signal transfer. It protects people and equipment, reduces ground loops, and handles large common-mode voltages. Isolation may be optical, magnetic, capacitive, transformer-based, or implemented with isolated DAQ modules.

Isolation example

A sensor mounted on equipment with a different ground potential may exceed the DAQ common-mode range. An isolated signal conditioner allows the sensor side and DAQ side to float relative to one another within the isolation rating.

4.6 Sensor excitation

Some sensors require external power or excitation:

- Strain gauges and load cells require bridge excitation.
- RTDs require a controlled current or voltage.
- IEPE accelerometers require constant-current excitation.
- Some pressure transducers require regulated DC supply.

Excitation stability directly affects measurement accuracy.

4.7 Bridge completion

Quarter-bridge strain gauges do not form a full Wheatstone bridge by themselves. Bridge completion adds precision resistors to complete the bridge. The completion network must be stable, low noise, and configured correctly for the gauge resistance.

4.8 Cold-junction compensation

Cold-junction compensation is signal conditioning specific to thermocouples. The DAQ must know the temperature at the reference junction, often where the thermocouple wire connects to copper terminals. Without compensation, the reported temperature can be biased by terminal-block temperature.

4.9 Linearization

Many sensors are nonlinear. Linearization converts raw voltage, current, or resistance into engineering units using equations, lookup tables, or calibration curves. Thermocouples, RTDs, and some flow meters commonly require linearization.

4.10 Current-to-voltage conversion

DAQ analog inputs usually measure voltage, so a 4 to 20 mA current loop often uses a precision shunt resistor:

$$V = IR.$$

With $R = 250\ \Omega$,

$$4\ \text{mA} \rightarrow 1\ \text{V}, \quad 20\ \text{mA} \rightarrow 5\ \text{V}.$$

Use precision resistors with suitable power rating and temperature coefficient.

4.11 Differential versus single-ended measurements

- **Single-ended:** signal measured relative to a common reference.
- **Differential:** voltage measured between two input terminals.

Differential measurements reject noise common to both leads and are preferred for low-level or long-cable signals.

$$V_{\text{meas}} = V_+ - V_-.$$

4.12 Common-mode voltage

Common-mode voltage is the voltage shared by both differential inputs relative to DAQ ground:

$$V_{\text{CM}} = \frac{V_+ + V_-}{2}.$$

Even if $V_+ - V_-$ is small, a large common-mode voltage can exceed the input specification and cause saturation or damage.

4.13 Common-mode rejection

Common-mode rejection ratio (CMRR) measures how well an amplifier rejects common noise:

$$\text{CMRR}_{dB} = 20 \log_{10} \left(\frac{A_d}{A_{cm}} \right),$$

where A_d is differential gain and A_{cm} is common-mode gain. Higher is better.

4.14 Input impedance

Input impedance is the effective resistance/capacitance seen by the signal source. Low input impedance can load the sensor and change the measured value. A voltage measurement system should usually have much higher input impedance than the source impedance.

4.15 Ground loops

A ground loop occurs when two or more ground paths create circulating current. This current can add noise or offset to low-level measurements. It is common when sensors, DAQ systems, power supplies, and test hardware are grounded at different points.

4.16 Shielding and cable routing

Shields intercept electric-field noise; twisted pairs reduce magnetic pickup; separation from high-current or switching cables reduces coupling. Shield termination strategy depends on frequency, equipment bonding, isolation, and manufacturer guidance.

5 Sampling, Timing, and ADCs

5.1 Sampling rate

Sampling rate f_s is samples per second. It must be high enough to capture the signal of interest, support control timing, and allow anti-alias filtering. A practical rule is often 5 to 10 times the highest frequency of interest for time-domain shape, but final selection depends on the objective.

5.2 Nyquist criterion

The theoretical minimum sampling rate for a maximum signal frequency $f_{\max}$ is:

$$f_s > 2f_{\max}.$$

In practice, choose a higher rate to allow filter roll-off and transient fidelity.

5.3 Aliasing equation

A sinusoid at frequency f sampled at f_s can appear at an alias frequency:

$$f_{\text{alias}} = |f - kf_s|,$$

for an integer k chosen so the result falls below $f_s/2$.

Aliasing example

If $f_s = 1000$ Hz and unwanted vibration is at 900 Hz, it may appear as a 100 Hz signal because 900 is 100 Hz away from 1000 Hz. Without an analog anti-alias filter, the data may falsely suggest a 100 Hz physical phenomenon.

5.4 ADC quantization

For an N -bit ADC with full-scale span V_{span} :

$$\Delta V = \frac{V_{\text{span}}}{2^N}.$$

If the input range is -10 to $+10$ V, then $V_{\text{span}} = 20$ V.

16-bit ADC example

$$\Delta V = \frac{20 \text{ V}}{65536} \approx 305 \text{ } \mu\text{V}.$$

If a sensor scaling is $300 \text{ } \psi/\text{V}$, the ideal pressure step is:

$$305 \text{ } \mu\text{V} \times 300 \frac{\psi}{\text{V}} \approx 0.0915 \text{ } \psi.$$

5.5 Effective number of bits

Noise reduces useful resolution. Effective number of bits (ENOB) describes actual performance after noise and distortion:

$$\text{ENOB} \approx \frac{\text{SNR}_{dB} - 1.76}{6.02}.$$

A 16-bit ADC may behave like a 13- or 14-bit system if noise is significant.

5.6 Synchronization across channels

Channels may be sampled simultaneously, multiplexed through one ADC, or acquired by separate devices. Time alignment depends on shared clocks, triggers, cable delays, filter delays, and timestamp strategy. For fast transients, hardware synchronization is preferable to software timestamps.

5.7 Deterministic versus non-deterministic acquisition

Deterministic timing means operations occur within known timing bounds. Hardware-timed DAQ and real-time controllers are more deterministic than normal desktop operating systems. Non-deterministic processes can still be acceptable for logging and user interfaces if they do not control safety-critical timing.

6 Grounding, Shielding, and Noise Troubleshooting

6.1 Common sources of noise

- 50/60 Hz mains pickup.
- Motor drives and switching supplies.
- Radio-frequency interference.
- Ground loops.
- Poor shield termination.
- Long high-impedance sensor leads.
- Mechanical vibration or real physical pulsation.
- ADC range or amplifier saturation.

6.2 Differential inputs and twisted pairs

Twisted pairs help ensure external magnetic fields induce nearly equal voltage in both conductors. A differential input then subtracts the two wires, rejecting common noise.

6.3 Single-point grounding

Single-point grounding connects signal references at one controlled point to avoid circulating currents. It is often useful for low-frequency systems, but it is not universal. At high frequencies, cable shields may need multi-point bonding to reduce RF impedance.

6.4 Shield termination

Possible strategies include one-end termination, both-end termination, capacitive termination, or chassis 360-degree termination. The correct choice depends on noise frequency, safety bonding, isolation, equipment instructions, and whether the shield carries fault current.

6.5 Troubleshooting method

Use a disciplined process:

1. Make the system safe.
2. Identify affected channels and conditions.
3. Look at raw data and spectrum if possible.
4. Correlate noise with pumps, valves, power supplies, or motion.
5. Inject known signals at successive points in the chain.
6. Change one variable at a time.
7. Document the result and root cause.

Question. A pressure measurement is stable when the pump is off but noisy when the pump starts.

Strong answer. Treat both physical and electrical causes as plausible. The pump could create real pressure ripple, mechanical vibration, electromagnetic interference, ground current, or signal-conditioner saturation. Compare nearby pressure sensors, review raw and filtered data, check frequency correlation with pump speed or drive switching, inspect cable routing, verify shield continuity, measure at the sensor and at the DAQ, and substitute a known signal. Do not filter away the signal until you know whether it is noise or real pressure behavior.

7 Controls and Automated Test Operations

7.1 Open-loop versus closed-loop control

Open-loop control commands an actuator without using measured feedback. Example: open a valve to 30 percent for 10 seconds.

Closed-loop control compares measured output to a setpoint and adjusts the actuator to reduce error.

$$e(t) = r(t) - y(t),$$

where $r(t)$ is setpoint and $y(t)$ is measured output.

7.2 PID control

A PID controller computes:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}.$$

- K_p : proportional response to current error.
- K_i : integral response to accumulated error.
- K_d : derivative response to rate of error change.

7.3 Overshoot and settling time

Overshoot is how far the response exceeds the target. Settling time is how long it takes to remain within a specified band around the setpoint. In pressure testing, excessive overshoot can violate limits; slow settling can waste test time or invalidate hold intervals.

7.4 Sensor delay

Delay adds phase lag and can destabilize feedback loops. If a pressure sensor, filter, or communication path delays the measurement, the controller may react to old information.

7.5 Actuator saturation

Saturation occurs when the controller asks for more actuator authority than is available, such as a valve command below 0 percent or above 100 percent. If the integrator continues accumulating error during saturation, windup occurs.

7.6 Integrator windup and anti-windup

Integrator windup causes overshoot and slow recovery after saturation. Anti-windup methods include clamping the integrator, back-calculation, conditional integration, and resetting during mode changes.

7.7 Rate limiting and setpoint ramping

Rate limiting constrains how quickly commands or setpoints change. Setpoint ramping prevents step commands from causing unsafe transients.

$$r_{k+1} = r_k + \text{clip}(r_{\text{target}} - r_k, -R\Delta t, R\Delta t).$$

7.8 Discrete-time implementation

Digital controllers update at sample time T_s . Integral and derivative terms must be implemented with correct timing:

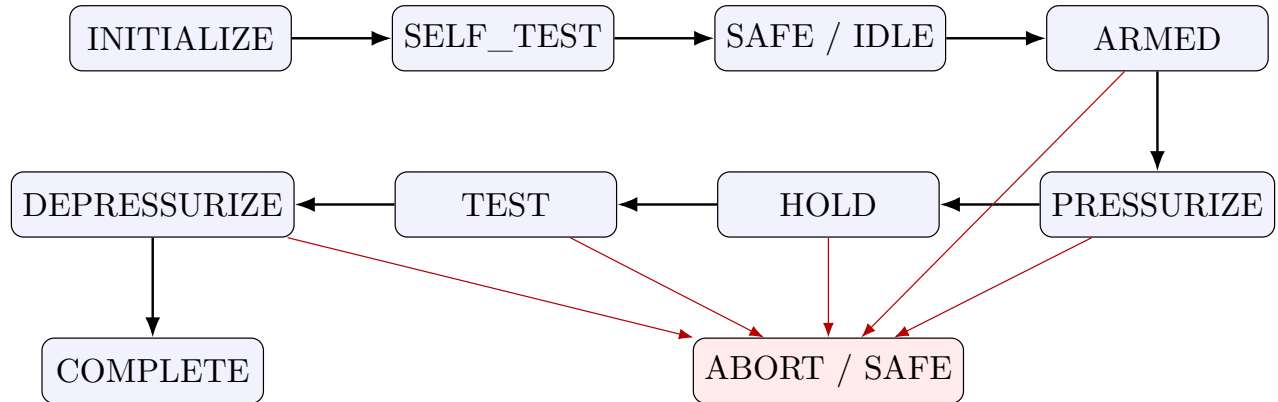
$$I_k = I_{k-1} + K_i e_k T_s,$$

$$D_k = K_d \frac{e_k - e_{k-1}}{T_s}.$$

Incorrect T_s is a common source of unstable control behavior.

7.9 State machines

Test automation should use explicit states rather than a long unstructured script. Each state has entry criteria, actions, exit criteria, timeouts, alarms, and abort transitions.



7.10 Permissives, interlocks, and watchdogs

- **Permissive:** condition that must be true before an action is allowed.
- **Interlock:** protective logic that prevents or stops unsafe operation.
- **Watchdog:** mechanism that detects software or communication failure.

7.11 Emergency shutdown and safe state

Safe state is system-specific. It may mean venting pressure, closing supply valves, opening relief paths, removing actuator power, or holding position. The safe state must be defined by hazard analysis, not by convenience.

Question. A pressure-control loop oscillates around setpoint. What could cause it?

Strong answer. Possible causes include excessive proportional or integral gain, process delay, sensor filtering, actuator deadband, valve stiction, saturation, incorrect loop timing, noisy measurement, poor operating point, or unit errors. Review setpoint, measurement, error, controller output, valve feedback, and saturation flags before retuning.

8 Test Safety and Fault Management

8.1 Layered safety

Inherently safer design

Physical limits and protective hardware

Independent interlocks

Control software protections

Procedural controls

Training and PPE

8.2 Fail-safe versus fail-operational

Fail-safe means the system transitions to a safer condition after a failure. **Fail-operational** means the system continues operating after a failure, often with redundancy. Test stands commonly prioritize fail-safe behavior unless stopping creates a greater hazard.

8.3 Fault detection

Fault detection can include limit checks, rate-of-change checks, sensor disagreement, stale data detection, range checks, communication timeouts, actuator feedback mismatch, and heartbeat/watchdog monitoring.

8.4 Fault response

Fault response should be defined before the test. Example responses:

- Alarm only.

- Controlled hold.
- Controlled depressurization.
- Immediate abort.
- Hardware trip independent of software.

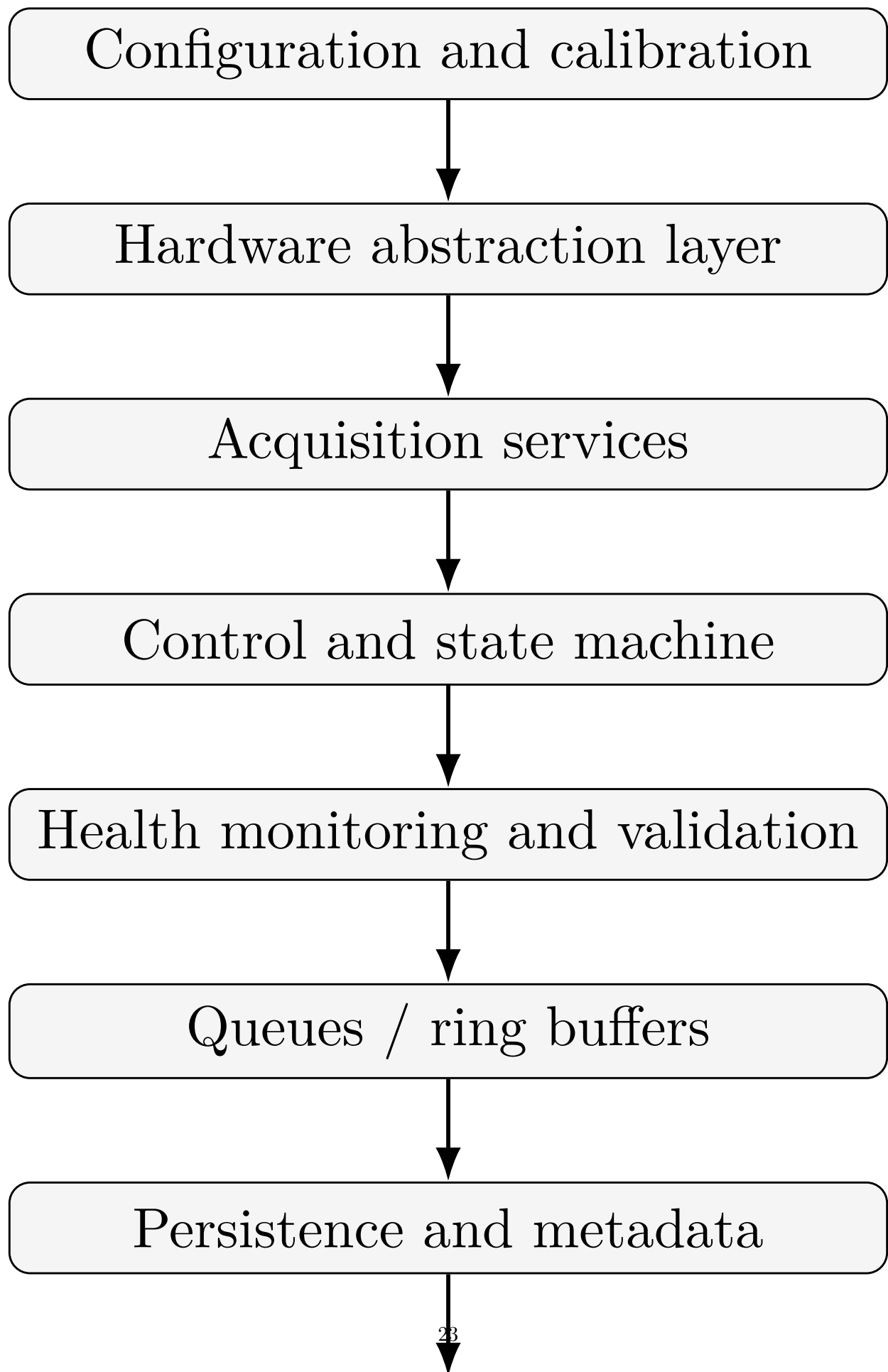
Question. What should the test stand do if communication with the controller is lost?

Strong answer. The response must come from hazard analysis. Do not assume that holding the last command is safe. Outputs should transition predictably on communication loss, watchdog timeout, application crash, or power loss. Critical protection should remain effective even when supervisory software is unavailable.

9 Python Software Engineering for DAQ and Controls

9.1 Separation of concerns

Good software separates hardware access, scaling, control logic, storage, user interface, and analysis.



9.2 Classes and modules

Use modules to group related functionality. Use classes for devices, channels, configurations, and services when they have state and behavior. Avoid a single large script that mixes hardware calls, scaling, plotting, and file writing.

9.3 Type hints

Type hints document intent and help tools detect mistakes:

```
def scale_pressure(voltage: float, slope: float, offset: float) -> float
```

They are especially useful in DAQ code because units and data shapes matter.

9.4 Exceptions

Exceptions should represent exceptional conditions that the caller can handle or surface. Do not silently catch all exceptions and continue as if data are valid. For test systems, silent failure can create false confidence.

9.5 Context managers

Context managers ensure resources are opened and closed correctly:

```
with open(path, "wb") as f:
```

For DAQ, the same concept applies to hardware tasks: configure, start, stop, and release resources deterministically.

9.6 Logging

Use structured logs for events, state transitions, commands, faults, software versions, configuration IDs, and operator actions. Logs should support root-cause analysis after a test.

9.7 Unit testing and mocking hardware

Unit tests verify scaling, state transitions, configuration validation, and analysis calculations. Mock hardware allows tests without physical equipment. However, mocked tests do not replace integration tests with real hardware.

9.8 Concurrency: threads, processes, queues, async

- **Threads:** useful for I/O-bound tasks such as reading drivers and writing files.
- **Processes:** useful for CPU-heavy analysis that should bypass the Python GIL.
- **Queues:** safe handoff between acquisition, storage, and UI tasks.
- **Async I/O:** useful for many network or file I/O operations, less useful for CPU-heavy work.

9.9 Bounded queues and backpressure

Unbounded queues can hide failures until memory is exhausted. Bounded queues force the design to define what happens if consumers fall behind.

Question. What happens if the disk writer falls behind?

Strong answer. Monitor queue depth and disk throughput. Warn early. Depending on criticality, abort safely before data loss, reduce nonessential visualization, or buffer for a bounded time. Do not silently drop samples unless that behavior is explicitly allowed, logged, and visible to users.

9.10 Configuration management

Configuration files should define channel names, hardware addresses, ranges, scaling coefficients, calibration IDs, sample rates, alarms, and units. Store a copy or hash of the configuration with each test dataset so analysis is reproducible.

10 NI Hardware, LabVIEW, and DAQmx

10.1 CompactDAQ versus PXI/PXIe

CompactDAQ is modular and convenient for distributed or smaller test systems. **PXI/PXIe** provides higher performance, tight synchronization, and many specialized modules in a chassis architecture.

10.2 Analog input and output

Analog input measures continuous signals such as voltage, current, temperature, pressure, strain, and acceleration. Analog output commands continuous values, such as valve position or actuator setpoint.

10.3 Digital I/O

Digital inputs read discrete states such as limit switches, valve feedback, interlock status, and emergency-stop contacts. Digital outputs command relays, solenoids, enables, or indicators.

10.4 Counter/timer channels

Counters measure frequency, period, encoder position, pulse width, and event counts. They are useful for flow meters, tachometers, encoders, and precise timing.

10.5 Hardware-timed acquisition

Hardware timing uses a sample clock on the DAQ device. The host program reads blocks of samples; it does not schedule each sample individually. This improves sample interval consistency and synchronization.

10.6 Software-timed acquisition

Software timing relies on the host operating system to decide when each read or write occurs. It is suitable for slow non-critical tasks but not for precise sampling or control timing.

10.7 Sample clocks and triggers

A sample clock defines when samples are taken. A start trigger begins a task. A reference trigger can capture data before and after an event. Shared clocks and triggers synchronize multiple modules or devices.

10.8 Buffered tasks

Buffered acquisition stores samples in hardware or driver buffers before the application reads them. Buffer sizing must handle expected latency without overflow.

10.9 Continuous versus finite sampling

Finite sampling collects a specified number of samples. Continuous sampling runs until stopped and requires the application to keep reading data fast enough.

10.10 LabVIEW producer-consumer architecture

A producer loop acquires data or handles events. A consumer loop processes, displays, or writes data. Queues decouple timing so slow consumers do not block time-critical producers.

10.11 LabVIEW state machines

LabVIEW state machines use cases or queued messages to transition between states. The same engineering rules apply: explicit states, defined transitions, timeouts, fault handling, and logging.

11 Data Processing, Validation, and Visualization

11.1 Raw data preservation

Never overwrite raw data with filtered or scaled-only values. Raw data allow reanalysis if calibration, filtering, or requirements change.

11.2 File integrity

Use file size checks, checksums, complete headers, finalization flags, and metadata validation to detect incomplete or corrupted files.

11.3 Engineering-unit conversion

Raw voltage or counts are converted to engineering units:

$$y = mx + b$$

for a linear sensor, or

$$y = f(x)$$

for a calibration curve. Always preserve units.

11.4 Invalid and missing samples

Invalid samples should be flagged, not silently filled. Missing data may be due to sensor dropout, saturated ADC, dropped packets, file corruption, or intentional test pauses.

11.5 Time alignment

Channels can be misaligned by separate clocks, multiplexing, filters, sensor response, communication delays, or timestamp interpretation. Alignment should be based on known timing architecture or identifiable physical events.

11.6 Filtering in analysis

Document filter type, order, cutoff, phase behavior, and whether it is causal or zero-phase. Avoid using filters to make data look better without engineering justification.

11.7 Metrics and pass/fail

A good report shows:

- Metric value and units.
- Acceptance limit.
- Margin to limit.
- Data-quality status.
- Time interval used.
- Calibration and software versions.

Question. How would you determine whether a component passed a test?

Strong answer. Define acceptance criteria with the responsible engineering authority. Identify exact channels and time intervals. Verify data quality. Compute metrics with units and uncertainty. Report result, limit, margin, quality flags, raw-data reference, configuration, calibration, and analysis-code version. If required data are invalid, report indeterminate rather than false pass or false fail.

12 Databases and Data Architecture

12.1 Raw waveform storage versus metadata storage

Large waveform data are often stored in files such as TDMS, HDF5, Parquet, or binary formats. Metadata are stored in a database for searchability. This avoids putting huge time-series arrays directly into relational tables while preserving traceability.

12.2 Relational databases

Relational databases store structured tables with relationships. Example tables: Tests, TestArticles, Channels, Calibrations, Configurations, Operators, and Results.

12.3 Time-series databases

Time-series databases optimize queries over timestamped data. They are useful for long-duration monitoring but may not replace immutable raw files for qualification evidence.

12.4 Schema design

A schema should answer: What was tested? When? By whom? With which procedure, software, DAQ configuration, calibration, hardware serial numbers, and acceptance limits?

12.5 Checksums

A checksum or hash detects file changes:

$$\text{hash} = H(\text{file bytes}).$$

Store hashes with metadata to detect corruption or unintended modification.

12.6 Calibration traceability

Each channel should link to the calibration record used during the test. If a calibration expires or is later found invalid, affected tests can be identified.

13 Electrical Schematics and Wiring Diagrams

13.1 Power distribution

Understand source, breakers, fuses, disconnects, power supplies, voltage rails, returns, and protective earth. Confirm voltage rating and current capacity.

13.2 Relays and contactors

Relays switch lower-power circuits; contactors switch higher-power loads. Coil voltage, contact rating, normally open/normally closed behavior, and flyback suppression matter.

13.3 Terminal blocks and connector pinouts

Terminal blocks provide wiring transition points. Connector pinouts identify signals, returns, shields, and spare pins. Wrong pinouts are common integration failures.

13.4 Normally open and normally closed contacts

A normally closed safety contact is closed in the normal safe condition and opens on trip or wire break. This can make some faults detectable.

13.5 Return paths

Every signal requires a return path. Missing or shared return paths can create offsets, noise, or unsafe current flow. Trace both signal and return in the schematic.

13.6 Isolation boundaries

Isolation boundaries show where signals cross between different electrical domains. Do not accidentally defeat isolation with shields, commons, or test equipment grounds.

Question. A sensor reads full scale regardless of applied input. How would you diagnose it?

Strong answer. Verify safety and determine whether the value is raw ADC saturation or software scaling. Check channel type, terminal mode, input range, scaling, excitation, and calibration. Trace the schematic from sensor to terminal block, conditioner, connector, and DAQ. Measure known points. Look for open return, short to excitation, incorrect pinout, bridge completion error, common-mode violation, or saturated conditioner. Apply a simulator at the DAQ to separate wiring/sensor from acquisition.

14 Test Stand Commissioning

14.1 Design and schematic review

Review hazards, requirements, drawings, channel list, I/O list, control philosophy, safe state, interlocks, procedures, and acceptance criteria before energizing hardware.

14.2 Continuity and isolation checks

Continuity verifies intended connections. Isolation verifies unintended connections do not exist. Perform these before power is applied.

14.3 Controlled power-up

Power up in stages, checking current draw and voltage rails. Use current limits, disconnect loads when appropriate, and stop if values are unexpected.

14.4 I/O checkout

For each input, apply a known stimulus and verify raw value, scaled engineering unit value, limits, metadata, and logging. For each output, command safely and verify independent feedback.

14.5 Fault injection

Deliberately simulate failures such as sensor open circuit, overpressure signal, communication loss, actuator feedback mismatch, and power loss. Verify the expected alarm or abort behavior.

14.6 Envelope expansion

Do not jump directly to full-energy testing. Use dry runs, low-energy tests, and incremental expansion while reviewing data and discrepancies.

15 Open-Ended System Design Answer

15.1 Clarify requirements

For a pressurized component test, ask:

- What component and failure modes are under test?
- What medium, pressure, temperature, and duration?
- What dynamic events must be captured?
- Which measurements are safety-critical?
- What must be controlled: valves, pumps, regulators, heaters?
- What is the safe state?
- What pass/fail criteria and uncertainty are required?
- What standards, reviews, and approvals apply?

15.2 Define instrumentation

Example channel set: inlet pressure, outlet pressure, differential pressure, temperature, flow rate, valve command, valve position, supply voltage/current, ambient conditions, vibration, and leak detection.

15.3 Define measurement chains

Physical quantity → Sensor → Excitation → Conditioning → DAQ → Scaling → Validation → Storage.

15.4 Define controls and safety

Use a state machine with permissives, controlled ramps, limits, timeouts, watchdogs, abort states, operator authority rules, and independent safety protections when required.

15.5 Define data architecture

Use hardware timestamps where possible. Store raw data, metadata, channel map, calibration versions, configuration versions, event logs, operator actions, software commit, and test article serial number.

15.6 Define verification

Verify with unit tests, simulated devices, electrical simulators, integration tests, fault injection, dry runs, low-energy tests, long-duration tests, and readiness review.

16 Behavioral Interview Preparation

16.1 Leadership principles

Blue Origin lists: Passion for Our Mission, Customer Focus, Deliver Results, Embrace Team Blue, Resourceful, Bias for Action, Ownership, Operational Excellence, Earn the Trust of Others, Hire and Develop the Best, Insist on the Highest Standards, Technical Ambition and Simplicity, Practice Humility, and Have Backbone; Disagree and Commit.

16.2 STAR format

Part	Share	What to include
Situation	10–15%	Context only. Do not over-explain background.
Task	10–15%	Your responsibility and desired outcome.
Action	55–65%	What you personally did, tradeoffs, risks, collaboration, validation.
Result	15–20%	Quantified outcome, lesson, and mechanism that prevented recurrence.

16.3 Ownership answer framework

Question. Tell me about a problem outside your formal responsibility that you took ownership of.

Strong answer. Describe a system-level issue that crossed team boundaries. Explain why you took ownership, how you mapped the end-to-end chain, who you involved, what data isolated root cause, what fix you implemented, how you validated it, and what mechanism prevented recurrence.

16.4 Safety/high-standards answer framework

Question. Tell me about a time you stopped or delayed work because of a quality concern.

Strong answer. Use a real example where proceeding would have risked invalid data, unsafe operation, or downstream rework. Show evidence-based escalation, stakeholder communication, correction, independent verification, and a mechanism that made future recurrence less likely.

16.5 Disagree and commit answer framework

Question. Tell me about a time you disagreed with a technical decision.

Strong answer. Show that you understood the other view, presented data, proposed a way to reduce uncertainty, and focused on mission risk rather than preference. After the decision, explain how you supported execution even if your preferred option was not selected.

17 Questions to Ask the Panel

17.1 Hiring manager

- What are the highest-priority test capabilities this engineer would support in the first six months?
- What differentiates an Engineer II who meets expectations from one who excels?
- Is the work mostly new test-system development, sustaining existing systems, or both?
- How is ownership divided among test engineering, instrumentation, controls, software, and operations?

17.2 DAQ/software engineer

- What DAQ platforms and software architectures are most common?
- How do you manage channel configuration, calibration, and metadata?
- How are software releases validated before use on hardware?
- How do you simulate hardware interfaces during development?

17.3 Controls/test operations engineer

- How are safety-critical protections divided among hardware interlocks, real-time control, and supervisory software?
- What level of automation is typical during component testing?
- What are the most common causes of commissioning delays?
- How are abort sequences validated?

18 Final Checklist

18.1 Technical checklist

- Sensor range and uncertainty.
- Pressure transducers.
- Thermocouples and CJC.
- RTDs and lead compensation.
- Load cells and bridge circuits.
- Amplification and attenuation.
- Anti-alias filters.
- Isolation and grounding.
- ADC resolution and ENOB.
- Sampling and synchronization.
- State machines and PID.
- Interlocks and watchdogs.
- Safe state and fault response.
- Python queues and logging.
- DAQmx clocks and triggers.
- Data traceability.
- Database metadata.
- Schematics and wiring.
- Commissioning sequence.
- Open-ended system design.

18.2 Closing statement

Recommended close

“Thank you for the discussion. This role strongly matches the work I want to continue developing—real-time systems, data acquisition, controls, test automation, and engineering analysis. I am particularly interested in owning the complete path from instrumented hardware to trusted test results. Based on what I have learned, I believe my background in [strongest two areas] would let me contribute while continuing to grow in [relevant area]. I am excited about the opportunity to support safe and repeatable testing for New Glenn.”